Question ID: b0543157

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$g(x) = \frac{x^2 - x - a}{x^3 - x - b}$$

The function g is defined by the given equation, where a and b are constants. In the *xy*-plane, the graph of $y = g(x)$ passes through the point $(0, 22)$, and $g(-22) = 0$. What is the value of b?

Answer

A. 23

B. 22

C. -22

D. -23

Correct Answer: A

Rationale

Choice A is correct. It's given that in the *xy*-plane, the graph of $y = g(x)$ passes through the point $(0, 22)$. It follows that $g(0) = 22$. Substituting 0 for x and 22 for $g(x)$ in the given equation yields $22 = \frac{0^2 - 0 - a}{0^3 - 0 - b}$, which is equivalent to $22 = \frac{-a}{-b}$, or $22 = \frac{a}{b}$. Multiplying each side of this equation by b yields $22b = a$. Substituting $22b$ for a in the equation defining the function g yields $g(x) = \frac{x^2 - x - 22b}{x^3 - x - b}$. It's given that $g(-22) = 0$. Substituting -22 for x and 0 for $g(x)$ in the equation $g(x) = \frac{x^2 - x - 22b}{x^3 - x - b}$ yields $0 = \frac{(-22)^2 - (-22) - 22b}{(-22)^3 - (-22) - b}$. Since a rational expression equals 0 if and only if its numerator is 0, it follows that $(-22)^2 - (-22) - 22b = 0$, or $22^2 + 22 - 22b = 0$. Dividing each side of this equation by 22 yields $22 + 1 - b = 0$, or $23 - b = 0$. Adding b to each side of this equation yields $23 = b$, so the value of b is 23.

Choice B is incorrect. This is the value of $g(0)$, not the value of b.

Choice C is incorrect. This is a value of x for which $g(x) = 0$, not the value of b.

Choice D is incorrect and may result from conceptual or calculation errors.